Practical 1: Installation of Arduino IDE and Blink — Built-in LED and External LED Interfaced with Arduino

A. Objective

- To install and configure the Arduino IDE on a computer for sketch development and board programming.
- To understand the structure of a basic Arduino sketch (setup() and loop() functions).
- To blink the Arduino Uno's built-in LED using digitalWrite() and pinMode().
- To wire and blink an external LED connected to a digital pin, reinforcing safe circuit-building practice.

B. Practical Outcome

On successful completion of this practical, the student will be able to:

- Install the Arduino IDE and correctly select the board and COM port for an Arduino Uno.
- Write, compile, and upload a basic Arduino sketch.
- Control a digital output pin to blink the onboard LED at a specified interval.
- Build a simple LED + resistor circuit on a breadboard and control it from an Arduino digital pin.

C. Expected Affective Domain

(Attitudes, safety habits, and professional practices these practical aims to instil, per Bloom's affective domain)

- Develops careful, methodical habits in software installation and configuration (verifying driver installation, port selection) before attempting hardware work.
- Builds confidence in independently troubleshooting basic upload errors (wrong board/port selected, missing drivers) rather than immediately seeking help.
- Instils safe circuit-building practice: checking polarity and using a current-limiting resistor before powering an LED, recognizing the cost of skipping this step.
- Encourages a habit of incremental testing — verifying the built-in LED works before moving to a more complex external circuit — a foundational engineering discipline reinforced throughout the course.

D. Resources Required

Category	Items
Hardware	Arduino Uno board, USB Type-A to Type-B cable, breadboard, 1 LED (5mm, any colour), 1 resistor (220Ω–330Ω), 2 jumper wires (male-male)
Software	Arduino IDE (version 2.x recommended, downloaded from arduino.cc/en/software), USB driver (usually auto-installed with IDE; CH340 driver needed for some clone boards)
Computer	Windows/macOS/Linux PC or laptop with at least one free USB port and administrator rights to install software
Reference	Arduino Uno pinout diagram (printed or projected)

E. Experimental Setup and Code

Part 1: Arduino IDE Installation

- Download the latest Arduino IDE installer for your operating system from the official site: arduino.cc/en/software.
- Run the installer and accept the default options, including installation of any USB drivers when prompted (required on Windows for the board to be recognized).
- Connect the Arduino Uno to the computer via USB cable. The board's power LED should light up immediately.
- Open the Arduino IDE. Go to Tools → Board and select Arduino board type, i.e. "Arduino Uno."/ "Arduino Mega" etc.
- Go to Tools → Port and select the COM port (Windows) or /dev/tty.* port (macOS/Linux) corresponding to the connected Arduino. If no port appears, recheck the USB cable and driver installation.

Part 2: Blink the Built-in LED

The Arduino Uno has a built-in LED connected internally to digital pin 13, requiring no external wiring.

```
//Write the program to blink an Internal LED
```

- Open this sketch in the Arduino IDE (or use File → Examples → 01.Basics → Blink as a ready-made reference).
- Click Verify (checkmark icon) to compile the sketch and check for errors.
- Click Upload (arrow icon) to flash the sketch to the board. Watch for the "Done uploading" message in the status bar.
- Confirm the onboard LED (marked 'L' next to pin 13) blinks on and off once per second.

Part 3: Wire and Blink an External LED
Circuit: Arduino digital pin 8 → 220Ω resistor → LED anode (long leg); LED cathode (short leg) → Arduino GND.

Component Pin	Connects To (Board Pin)	Notes
LED Anode (long leg)	Digital Pin 8 (via 220Ω resistor)	Resistor limits current to protect the LED
LED Cathode (short leg)	GND	Completes the circuit

//Draw the circuit diagram of LED interfaced with Arduino board.

```
//Write the code to Blink an External LED interfaced with Arduino
```

- Build the circuit on the breadboard exactly as per the wiring table above, double-checking LED polarity (long leg = anode = positive) before powering on.
- Upload the Part 3 sketch and confirm the external LED blinks twice as fast as the built-in LED did in Part 2 (0.5s vs 1s interval).
- Optional extension: modify the delay() values to make the LED blink faster or slower, and try connecting a second LED to a different digital pin to blink alternately with the first.

F. Outcome / Observation

Students should record the following in their lab journal/observation sheet:

G. MCQs for the Practical

Q1. Which Arduino function runs only once when the board is powered on or reset?

A) loop()
B) setup()
C) begin()
D) init()

Q2. Which built-in constant refers to the Arduino Uno's onboard LED pin?

A) LED_PIN
B) ONBOARD_LED
C) LED_BUILTIN
D) PIN_LED

Q3. What is the purpose of the resistor in series with the external LED?

A) To increase the LED's brightness
B) To limit current and protect the LED from damage
C) To allow the LED to blink automatically
D) To convert digital signal to analog

Q4. Which function is used to set a pin as an output in setup()?

A) digitalWrite()
B) pinMode()
C) analogWrite()
D) setOutput()

Q5. In the external LED circuit, which leg of the LED should connect to GND?

A) The longer leg (anode)
B) The shorter leg (cathode)
C) Either leg, polarity doesn't matter
D) Neither — GND connects to the resistor only

Q6. What does digitalWrite(ledPin, HIGH) do?

A) Reads the current state of the pin
B) Sets the pin's output voltage to approximately 5V, turning the LED on
C) Sets the pin to input mode
D) Increases the pin's PWM duty cycle

Q7. If the Arduino IDE cannot find the correct COM port after connecting the board, what is the most likely first troubleshooting step?

A) Reinstall the operating system
B) Check the USB cable/connection and verify the board driver is installed
C) Increase the delay() values in the sketch
D) Change the LED's resistor value

Q8. What is the function of delay(1000) in the Blink sketch?

A) Pauses program execution for 1000 microseconds
B) Pauses program execution for 1000 milliseconds (1 second)
C) Sets the LED brightness to 1000
D) Repeats the loop 1000 times